



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY RAICHUR
MATHEMATICS CLUB

PROBLEM NUMBER: WP2505

DATE: DECEMBER 15, 2025

DUE DATE: DECEMBER 31, 2025

WEEKLY PROBLEM

EXERCISE. Let a and b be positive integers such that

$$\frac{a^2 + b^2}{ab + 1} = k$$

is an integer. Prove that k must be a perfect square.

Hints:

- First, rewrite the equation as a quadratic in a :

$$a^2 - kba + (b^2 - k) = 0$$

- If a is a root, let a' be the other root. Note that $a + a' = kb$ and $a \cdot a' = b^2 - k$.
- If a is one integer root of this quadratic, what can you say about the other root?
- Try choosing a solution with the smallest possible $a + b$, and see what that forces.

Examples:

- For $a = 1, b = 2$: $\frac{1^2 + 2^2}{1(2) + 1} = \frac{5}{3} \notin \mathbb{Z}$.
- For $a = 2, b = 2$: $\frac{2^2 + 2^2}{2(2) + 1} = \frac{8}{5} \notin \mathbb{Z}$.

The problem is asking you to understand why, whenever it is an integer, the quotient is forced to be a square — no matter what a and b are.



(Scan this to submit solution!)

Top Submissions for last Week's Problem: